

24 August 2026 14:42

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graph LR
    Clk[CLK] --> Counter[Counter]
    CompOut[Comparator Output] --> Counter
    Counter --> LED[LED controller frequency select]
    Counter --> Alarm[Alarm]

```

- Clock - Falling edge or otherwise need an inverter
- Low power
- CMOS
- Must have an enable

Cnumber = 00 ✓
 LED amb. freq. select = Low
 Alarm = Low

If $\text{Csubter} \neq 11$

If Comparison = High
Increment Counter

If $\text{Gunder} \neq 00$

If $\text{comparator} = \text{Low}$
Decrement counter

If $G_{\text{inter}} \neq 00$

LED sub. freq. select = High

IF counter = 11

Alarm = High

If number = 00

A form = Low

Else

Alarm unchanged

[illegible]

either make the comparator inverted (unsafe = low, safe = high)
or use an inverter

- Assuming inverted

a_A	a_B	Cam	$\overline{CTEN}$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

Q₃ Sum

Q ₄	00	01	11	10
0	0 ⁰	1 ¹	0 ³	0 ²
1	0 ⁴	0 ⁵	0 ⁷	1 ⁶

$$\overline{CTEN} = \overline{Q_A} \overline{Q_B} C_{in} + \underline{Q_A Q_B} \underline{C_{in}}$$